

Transport Situation Cards

Cell Transport & Osmosis Review

Instructions

For each scenario (#1-20), write what **type of transport** (passive or active) and then specify the **specific form of transport** (diffusion, exocytosis, endocytosis [specifically pinocytosis], etc.).

If it is an **osmotic solution**, specify the **movement of water in or out** of the cell and identify whether the cell is is a(n) **isotonic, hypotonic, or hypertonic solution**. **Tip: Try to draw out the osmotic scenario and where the water is moving.**

Scenario #	Type of Transport (A or P)	Form of Transport - be as specific as you can e.g. "endocytosis (phagocytosis)"	Water moving in or out? Hypo/hyper, or isotonic solution? (OSMOSIS ONLY)
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Scenario #	Type of Transport (A or P)	Form of Transport - be as specific as you can e.g. "endocytosis (phagocytosis)"	Water moving in or out? Hypo/hyper, or isotonic solution? (OSMOSIS ONLY)
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			

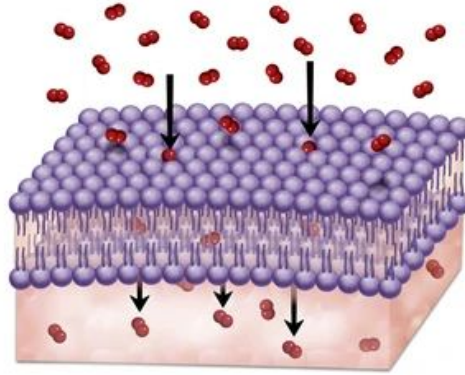
Scenario #1

A tsunami causes sea **salt water** to wash into a garden bed and the **plants wilt and die**.



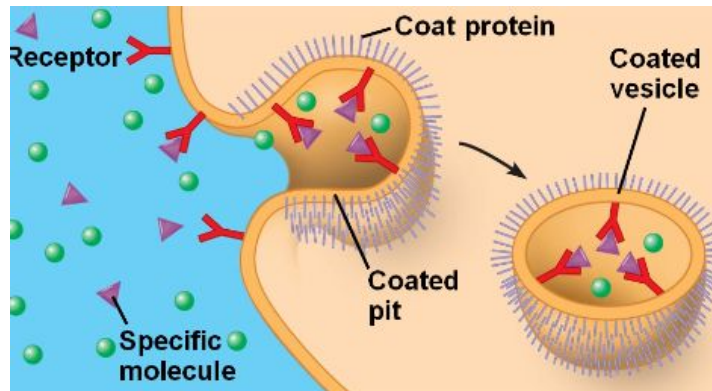
Scenario #2

The **movement of O_2 molecules** into and out of cells during respiration.



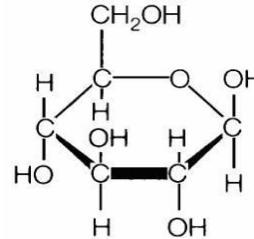
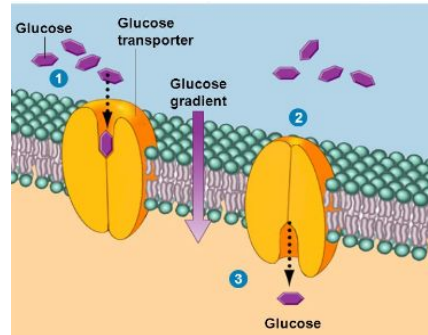
Scenario #3

Low-density lipoproteins outside of the cell bind to **receptor proteins** on the cell membrane. They are then **taken into the cell** through specialized **coated pits** that pinch off inside the cell.



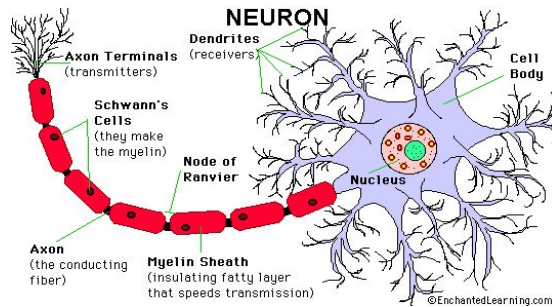
Scenario #4

Glucose molecules move across the membrane to supply the cell with energy. They must move through a **carrier** or **channel protein** embedded in the cell membrane because they are too large to go straight through the membrane.



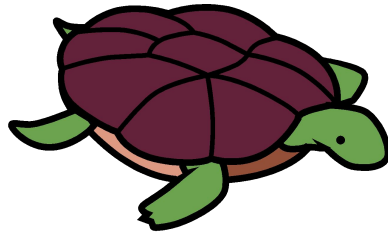
Scenario #5

Nerve cells **secrete neurotransmitters** in order to **send** signals to other neurons. To *secrete* means to *produce, release, or ooze*.



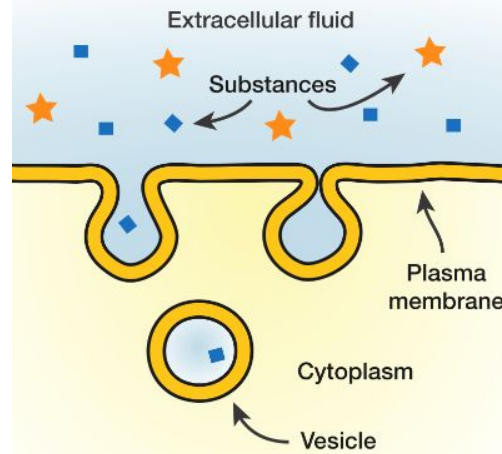
Scenario #6

What would happen if you took a **sea turtle (ocean)** and put it into **Lake Lanier (freshwater)**?



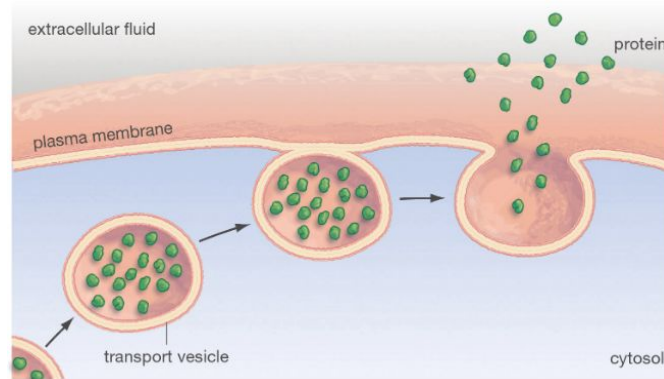
Scenario #7

Cells **take in** small amounts of solute and **liquid particles** by forming **vesicles** around them.



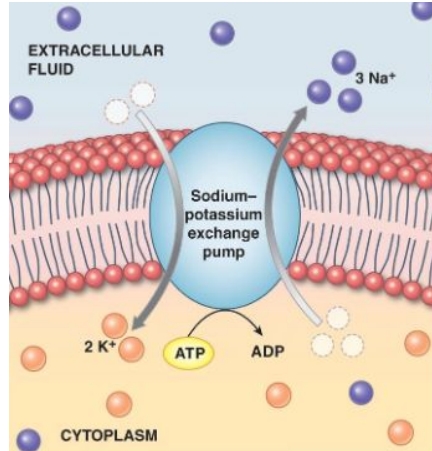
Scenario #8

Plant cells **secreting proteins** and carbohydrates via **vesicles** to build cell walls. To *secrete* means to *produce, release, or ooze*.



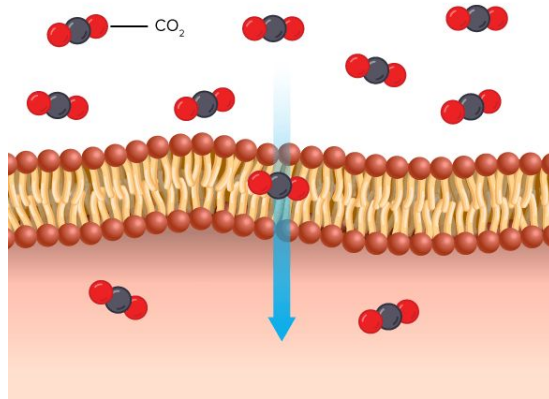
Scenario #9

2 potassium ions (K^+) are moved **into the cell while
3 sodium ions (Na^+) are moved **out** of the cell using
a **protein pump**.**



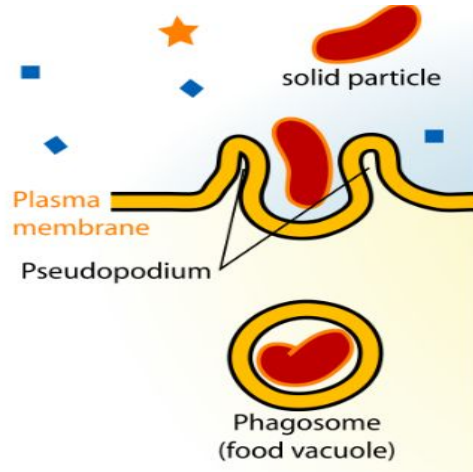
Scenario #10

The movement of CO_2 molecules into and out of cells during respiration.



Scenario #11

A **foreign cell** (like a **bacterium**) is **engulfed** by a white blood cell as an immune response. To *engulf* means to *surround* or *envelop*.



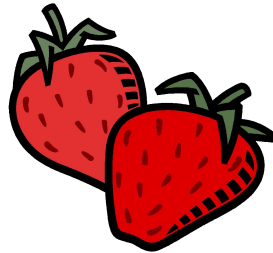
Scenario #12

Someone **sprays cologne** in the corner of the room;
eventually everyone in the room can smell it.



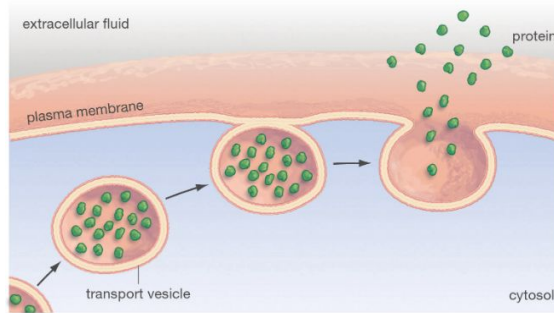
Scenario #13

I put **table sugar on strawberries** and put them in the fridge overnight. When I looked the next morning, I saw that they were sitting in a **pool of juice**. What happened to the strawberries?



Scenario #14

The pancreas produces **insulin** (a hormone). This large molecule **leaves the cell** and enters the **bloodstream via a vesicle**.



Scenario #15

Someone bakes cookies and the **smell travels** throughout the house from the kitchen.



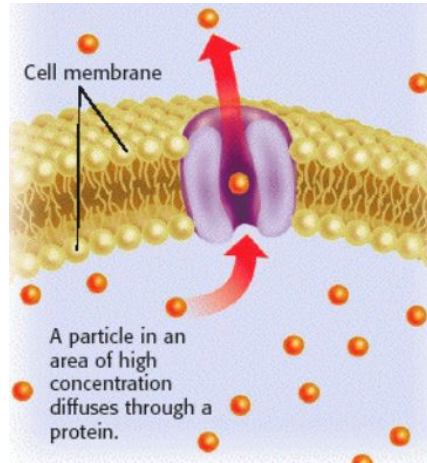
Scenario #16

Why do you get **thirsty** after eating **salty potato chips**?



Scenario #17

The movement of solutes **down** (“with”) the **concentration gradient** with the assistance of **channel proteins**.



Scenario #18

You leave a strawberry in a bowl of water. Eventually, the strawberry becomes **plump** and **swollen**. Overnight, the cell walls of the strawberry weaken, and the fruit becomes **soft** and **mushy**.

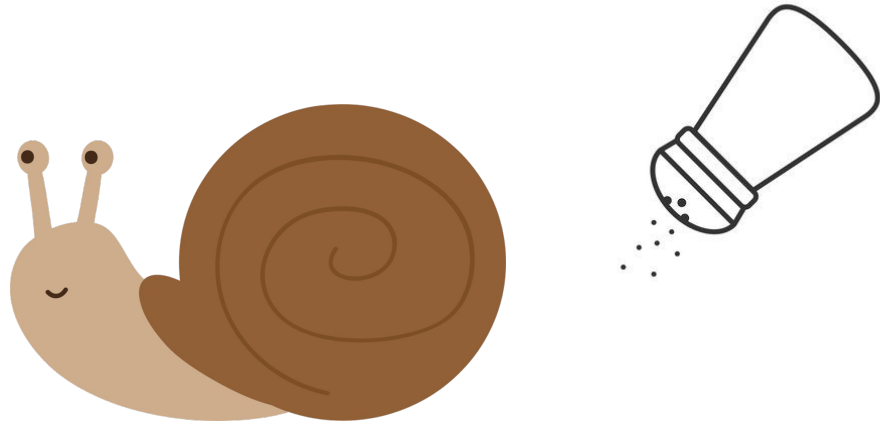
Scenario #19

Drinking **salty seawater** and then becoming **very sick** afterward.



Scenario #20

Snails can absorb water through their skin. What would happen if you **poured salt onto a snail**?



Warning: Use the next slide to check your answers!

#	Type of Transport	Form of Transport	Solution type (OSMOSIS ONLY)	What happens to the cell? (OSMOSIS ONLY)
Example	Passive	Diffusion	n/a	n/a
1	Passive	osmosis	hypertonic	cells shrink
2	Passive	simple diffusion	—	—
3	Active	receptor mediated endocytosis	—	—
4	Passive	facilitated diffusion	—	—
5	Active	exocytosis	—	—
6	Passive	osmosis	hypotonic	cells swell
7	Active	endocytosis (pinocytosis)	—	—
8	Active	exocytosis	—	—
9	Active	sodium-potassium pump	—	—
10	Passive	simple diffusion	—	—
11	Active	endocytosis (phagocytosis)	—	—
12	Passive	diffusion	—	—
13	Passive	osmosis	hypertonic	cells shrink
14	Active	exocytosis	—	—
15	Passive	diffusion	—	—
16	Passive	osmosis	hypertonic	cells shrink
17	Passive	facilitated diffusion	—	—
18	Passive	osmosis	hypotonic	cells swell
19	Passive	osmosis	hypertonic	cells shrink
20	Passive	osmosis	hypertonic	cells shrink